

Confidence and Stress Indicators

Overview

The Day 36 implementation introduces a Confidence and Stress Indicator Analyzer for the Zecpath AI System. The analyzer evaluates observable communication patterns in candidate responses and produces structured indicators related to behavioral confidence and possible stress signals.

This module extends the existing ConfidenceSentimentEngine instead of duplicating its functionality. The existing engine continues to analyze hesitation, response length, response pace, sentiment, uncertainty, and lexical consistency. The Day 36 analyzer adds additional observable response signals and combines them into a behavioral confidence assessment.

The output is designed to support interview analysis and communication evaluation. It does not provide a psychological diagnosis or determine a person's actual mental or emotional state.

Objectives

The main objectives of this implementation are:

- Detect textual long-pause indicators.
 - Detect immediately repeated words.
 - Identify configured stress-related terms.
 - Reuse the existing confidence and sentiment analysis engine.
 - Generate an observable behavioral confidence score.
 - Classify behavioral confidence into meaningful levels.
 - Produce structured and testable output.
 - Maintain compatibility with the existing Zecpath AI System.
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Architecture

The Day 36 implementation introduces the following module:

`scoring/confidence_stress_analyzer.py`

The primary class is:

`ConfidenceStressAnalyzer`

The analyzer internally uses:

`ConfidenceSentimentEngine`

This design avoids creating a duplicate confidence and sentiment system. Instead, the new analyzer extends the existing communication signal analysis with additional Day 36 indicators.

The analysis pipeline is:

Candidate Response | v ConfidenceSentimentEngine | +--> Hesitation +--> Response Length +--> Pace +--> Sentiment +--> Uncertainty +--> Consistency | v ConfidenceStressAnalyzer | +--> Long Pauses +--> Repeated Words +--> Stress Indicators | v Behavioral Confidence Score | v Confidence Level + Stress Level

Long Pause Detection

Long pauses are detected from textual indicators that may represent pauses or interruptions in a written or transcribed response.

The analyzer currently detects:

- Ellipsis sequences such as . . .
- Repeated dash sequences such as --
- Em dash pause indicators such as —

The analyzer records:

- Total pause count
- Ellipsis count
- Dash pause count
- Pause score

The pause score starts at 100 and decreases according to the number of detected pause indicators.

This is an observable text-based signal only. It does not directly measure real spoken pauses unless the response text or transcript contains corresponding indicators.

Repeated Word Detection

The analyzer identifies immediately repeated words in normalized response text.

For example:

I I worked on a project.

The repeated word I is detected.

The analyzer returns:

- Repetition count
- List of repeated words
- Repetition score

The repetition score starts at 100 and decreases when immediately repeated words are detected.

This signal can help identify possible hesitation or disruption in communication flow, but repeated words alone should not be interpreted as a psychological or diagnostic indicator.

Stress Indicators

The analyzer checks candidate responses for configured stress-related terms.

Current examples include:

- stressed
- stress
- nervous
- anxious
- worried
- overwhelmed
- pressure
- panic
- afraid
- fear
- frustrated
- confused

The analyzer produces:

- Stress term count
- Detected stress-related terms
- Stress score
- Stress level

Stress levels are classified as:

- Low
- Moderate
- High

The result represents lexical indicators present in the response. It does not determine whether a candidate is clinically stressed or diagnose any emotional or psychological condition.

Behavioral Confidence Score

The behavioral confidence score combines the existing confidence analysis with the new observable Day 36 indicators.

The current weighted components are:

- Existing confidence score: 50%
- Pause score: 15%
- Repetition score: 15%
- Stress score: 20%

The final score is constrained to the range:

0 to 100

The score is classified as:

- Strong: 75 and above
- Moderate: 50 to 74.99
- Needs Improvement: Below 50

The score represents observable communication behavior and response signals rather than an assessment of personality, intelligence, mental health, or psychological confidence.

Structured Output

The analyzer returns structured output containing:

- Raw response text
- Behavioral confidence score
- Behavioral confidence level
- Stress level
- Existing confidence analysis
- Long-pause analysis
- Repeated-word analysis
- Stress analysis

This structured format makes the analyzer suitable for integration with other interview and scoring components in the Zecpath AI System.

Testing

The Day 36 test suite validates:

- Analyzer initialization
- Long pause detection
- Dash pause detection
- Repeated word detection
- Responses without repeated words
- Stress term detection
- Low stress classification
- Behavioral confidence score range
- Behavioral confidence level classification
- Complete structured output
- Stress indicators during complete analysis

All Day 36 module tests passed successfully:

11 passed

The complete project regression suite also passed:

168 passed

Two warnings were reported during the full regression run. These warnings were related to existing dependency and test behavior and did not represent failures caused by the Day 36 implementation.

Limitations and Responsible Use

The ConfidenceStressAnalyzer evaluates observable textual communication signals only.

It should not be used to:

- Diagnose stress or anxiety.
- Make mental health conclusions.
- Infer personality traits.
- Determine a person's true emotional state.
- Make hiring decisions based only on stress-related terms or confidence scores.

The output should be treated as one supporting communication signal among multiple evaluation factors. Human review and appropriate fairness considerations remain important when using automated candidate analysis.

Conclusion

Day 36 extends the Zecpath AI System with a structured Confidence and Stress Indicator Analyzer.

The implementation builds on the existing ConfidenceSentimentEngine and adds:

- Long-pause detection
- Repeated-word detection
- Stress-related lexical indicators
- Behavioral confidence scoring
- Confidence level classification
- Structured integration-ready output

The module was validated with dedicated tests and the complete project regression suite, maintaining compatibility with the existing Zecpath AI System.